

An experimental protocol for video recording with a smartphone for use in a pre-trained neural network.

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Abstract — Parkinson's disease is a neurodegenerative disease that has, as one of its main symptoms, a change in the response of the individual's muscle activity. It is the second most common neurodegenerative disease, the first being Alzheimer's disease. Due to motor alterations being one of its main symptoms, the proposed protocol is intended to capture movements from a smartphone camera for further analysis in object recognition software (DeepLabCut) and finally extract important data that can help in the detection of the disease or its stage classification. The protocol was designed to be simple and of easy replicability; to do so, a simple test was used for the volunteer to exercise (drawing of spiral formats and sinusoidal waves), and to record it, a support was developed for the smartphone and for the sheet, thus avoiding many measurements and bad positioning according to the replication. A neural network pre-trained by the authors has been used in the recognition software, from which data will be extracted when the protocol is applied. This is an experimental protocol that has not yet undergone a pilot test, so points of improvement can be found. However, it's a protocol that can enable quick and practical tests for Parkinson's disease diagnosis and quantification.

Keywords — machine learning, medical diagnosis, computer vision, DeepLabCut, object tracking, video recording.

I. INTRODUCTION

Parkinson's disease (PD) was initially observed and characterized by James Parkinson in 1817 as a neurological disorder [1]. While some earlier studies described specific symptoms of this disease, they failed to establish correlations between them, treating the symptoms separately without linking them to a new syndrome [2]. Notably, Charcot's research contributed significantly to our understanding of the disease. He was the first to name the syndrome "Parkinson's disease" in honor of James Parkinson [3] and defined the four cardinal symptoms of the disease – tremor, muscular rigidity, bradykinesia, and postural instability – which remain critical in differentiating PD from other neurological syndromes, and he also pioneered the treatment of this pathology [4,5].

In addition to the motor symptoms listed by Charcot, PD can present non-motor symptoms, such as cognitive or neurobehavioral abnormalities, sleep disorders, pain, depression, anxiety, anosmia, among others [2,6]. While tremor is the most well-known motor sign, affecting over 70% of individuals with the disease, alterations in gait and

rigidity in the limbs are the most debilitating factors of PD [7]. Compromised mobility in individuals affected by PD can lead to difficulties in performing daily activities, significantly reducing their quality of life [8,9].

Despite the exact cause of PD not being fully understood, age remains the main risk factor, leading to an expected increase in the disease's prevalence due to population aging in the coming years [5,10]. PD is estimated to affect approximately ten million people worldwide [11], ranking as the second most common neurological disorder after Alzheimer's disease [10].

The diagnosis of PD relies on clinical analysis, considering if the patient presents at least two of the four cardinal symptoms and evaluating the individual's medical and family history. The Unified Parkinson's Disease Rating Scale (UPDRS), optimized by the Movement Disorder Society (MDS) as MDS-UPDRS, serves as the gold standard assessment tool, evaluating disease progression on a scale from zero (normal) to four (incapable of performing the task). However, this evaluation is subjective and prone to errors [12,13]. Consequently, significant efforts have been made to develop objective and quantitative methods for assessing the cardinal symptoms of PD and complementing UPDRS results [12,14,15].

To ensure better quality of life for individuals with PD and define personalized treatment plans, continuous monitoring of the disease is crucial. Regular consultations and evaluations are of utmost importance, and smart devices utilizing inertial sensors or providing voice and video recordings to physicians have been explored to offer greater comfort to patients [15–17]. Consequently, data analysis obtained through these approaches and physiotherapeutic procedures can provide valuable information about symptom progression and the stage of PD.

In this context, tools that add objectivity to subjective methods like UPDRS or assess the progress of physiotherapeutic procedures are highly relevant for diagnosis, treatment, and PD monitoring. Thus, this project's focus is on markerless computer vision (CV) techniques, encompassing image and video analysis alongside artificial intelligence and machine learning algorithms to extract pertinent information that can assist in PD-related processes [18–23].

Computer vision involves using computational techniques to interpret and analyze videos or images and extract information such as angles, velocity, or acceleration of specific joints [19,23]. Concerning the use of CV techniques to aid in PD diagnosis without markers, [24] and [21] utilized handwriting data to feed machine learning algorithms, achieving accuracies of 67% and 98.45%, respectively.

Two noteworthy markerless computer vision tools, DeepLabCut (DLC), and MediaPipe (MP), both freely available and licensed, are highlighted. In this study, DLC will be the primary focus. DLC consists of several pre-trained deep residual neural networks, known as ResNets, from the ImageNet database. Alongside ResNets, instead of a classification layer at the network's output, deconvolutional networks are utilized to produce spatial probability densities. For each analyzed body part, these probability densities represent evidence of the body part's position [25].

In the realm of proprioception studies, recent research has explored innovative approaches to isolate and comprehend proprioceptive inputs. For instance, [26] designed a set of real-world proprioceptive tasks to classify or reconstruct Latin alphabet characters based on passive arm movement. By utilizing a dataset of pen-tip trajectories, this work computationally separated proprioception from active motion, offering insights into the unique role of proprioceptive cues in motor control and to create this they used a dataset of pen-tip trajectories for the 20 characters that can be handwritten in a single stroke, so they excluded the characters *f*, *i*, *j*, *k*, *t*, and *x*, which are obtained by multi-stroke [11].

Considering the above, this study addresses the central problem of how features derived from data obtained through markerless computer vision tools, in this case utilizing the DLC tool, can provide more objectivity to the monitoring of PD, correlate values to the stage of the pathology, and potentially assist in the diagnosis due to this correlation. This validation of VC tools without markers for PD monitoring will contribute to achieving the proposed objectives.

I. MATERIALS AND METHODS

This section provides comprehensive details about experimental protocol, including developed equipment, positioning of equipment for video recording, data preparation and data extraction from videos.

The protocol consists of capturing videos of volunteers who will draw a sequence of five spirals and, subsequently, five sine waves. To this end, the authors developed specific devices to support future data collection, having as backup some studies that captured images or videos with or without markers to be analyzed [19,27].

Considering some studies that have captured videos and images, the protocol was developed, selecting some features that can help create a more precise protocol for the experiment.

The perception was to take a lot of care with the camera and its positioning because everything depends on good image or video quality [28–31]. Considering this, the protocol was designed to be careful with the positioning of the camera and the conditions to have a greater quality on the videos.

For this protocol, a stand was designed to support the smartphone, which will be the camera, and an external mold for a paper sheet was developed. Those will allow positioning everything always right and be easier to replicate, since it will not be necessary to measure a lot of distances to get the right position, avoiding parallax errors and other errors that can occur [19,29,30,32].

The field of view was also considered; this can be important in the future when the data will be analyzed by the software that will track the pen tip. [33] showed that on the edges of the field of view, the software creates more errors or tends to be less precise. This problem is avoided with the positioning of the paper sheet, which will not have the draw spaces touching the edge of the field of view of the camera; those spaces are positioned closely to the center of the camera, which provides a more reliable result.

Some other things to be considered are having a well-lit room, a flat and stable table, and avoiding objects that won't be captured as desired objects (everything except the pen, the paper sheet, and the hand of the volunteer) in the field of view of the camera. Those are the basics for keeping a good environment to take the videos.

Just to maintain the quality of the videos, some basic care with the camera is needed, such as a good and clean lens, good lighting in the room, avoiding shades, and avoiding shaking the camera or changing its position during the video taking.

About the paper sheet and its content, it was discussed by the authors what the position of the content should be and what should be included based on the experience of the group with data collection and the study about the field of view. This was done to make the paper sheet as simple as possible and to have enough information for both volunteers and authors.

II. RESULTS

A. Smartphone stand

A stand (gadget) was developed to support an specific Apple smartphone, the iPhone 13 Pro Max. The stand has the exact same dimensions as the smartphone. This was done to avoid mistakes in positioning during the tests.

The gadget stands the smartphone perfectly, maintaining the same angle and the same distance from the paper sheet which the volunteers will use to draw.

This will facilitate the process while the data is collected several times. This gadget was designed by the authors using CAD software and then 3D printed with PLA

material. This provides good resistance and lowers shaking during the video recording.

Another benefit of the gadget is that the authors included an external mold to fit the paper sheet perfectly, avoiding malposition of the paper sheet. It allows that data collection to be faster and more reliable every time it is needed.

The figure one will show the prototype of the smartphone support with its external mold.

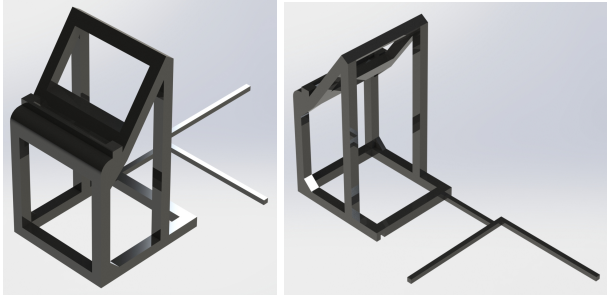


Figure 1. Smartphone gadget.

The paper sheet, which the future volunteers will use to draw, includes some instructions for the volunteers to follow. Including the samples of spire and sinus waves, as well as delimited spaces to make the sequence of five drawings for each form.

On this same sheet are printed the number of volunteers, their predominant hand, and the date of the collection.

B. Workflow for recording videos

First of all, it's necessary to print the specific sheet that was developed. Print as many as needed to collect the data. Once the printing is done, it's time to position the smartphone gadget on a flat and very stable table.

The gadget can be positioned at a distance that provides more comfort to the volunteer draw, but the sheet and the gadget need to be parallel with the table edge, like in the figure 2.

After having the gadget and the paper sheet parallel to the table edge, the paper sheet must be positioned on the external mold. This step can't be skipped; otherwise, the data collection will not have a good pattern. Once the paper sheet is positioned correctly, it's necessary to use some tape to ensure that the paper doesn't move away while the volunteer draws. Figure 3 will show exactly how it should be positioned.

After positioning the gadget and the paper sheet correctly, it's necessary to position the smartphone. To position the smartphone, remove any protection case; otherwise, some errors will be created. As the gadget was created specifically for this smartphone model, it cannot fit any other model. It is shown in figure 3. Meanwhile, having this model ensures that no adjustment is needed, or any special position is needed.

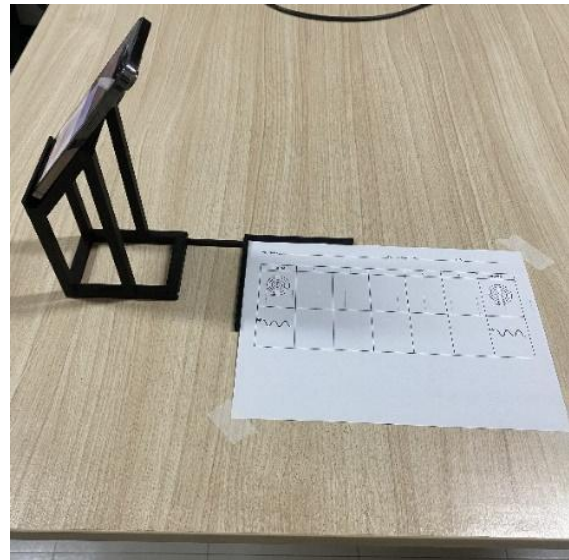


Figure 2. Equipment parallel with the edge of the table.

For the final adjustment, it's necessary to always use the same pen because the neural network is trained with this same pen. Using another type or color of pen will make the process more difficult, less reliable, or even impossible. With everything set, the volunteer can sit in a comfortable chair and get ready to draw a sequence of five draws.

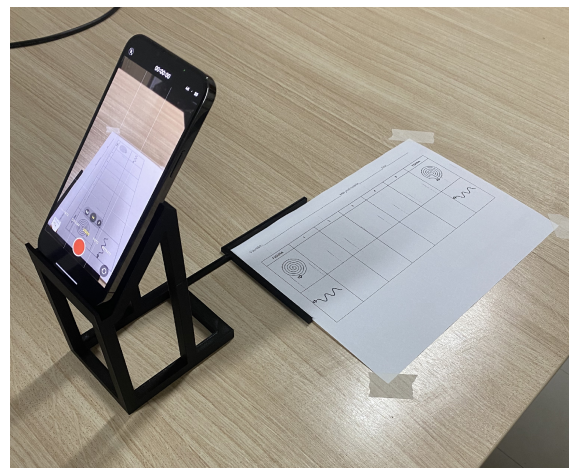


Figure 3. Equipment ready to start recordings.

With everything in its place, it's time to open the cellphone camera and verify if it's working well. If something is making the image blurry, it is necessary to remove the cellphone and get it cleaned. If there is a lot of shade in the pen or the camera cannot show the paper sheet and the pen, consider turning on the smartphone flashlight or some other lights. It's not recommended to capture videos with low illumination.

Once everything is working well, one person needs to be responsible for operating the camera. It's desirable that the video starts closely with the drawing. To facilitate this process, the same person who is responsible for the camera can give the command to begin the drawing to the volunteer at the same time he touches the button to start the video recording.

The video should last at least as long as necessary until the end of the last drawing of the sequence. Another video should be started to capture the other drawing sequence. So, one video will have the spiral drawing, and another video will have the sinus wave drawings.

In the drawing process, the volunteer should be incentivized to draw as fast and well as they can, but they can take as long as they need.

In the video recording process, the responsible subject has to be focused because it's necessary to change the cellphone focus to the box of the current drawing. If it isn't done, the camera will lose focus on the drawing, resulting in a bad video. Figure 4 shows how the screen should look while taking recordings.

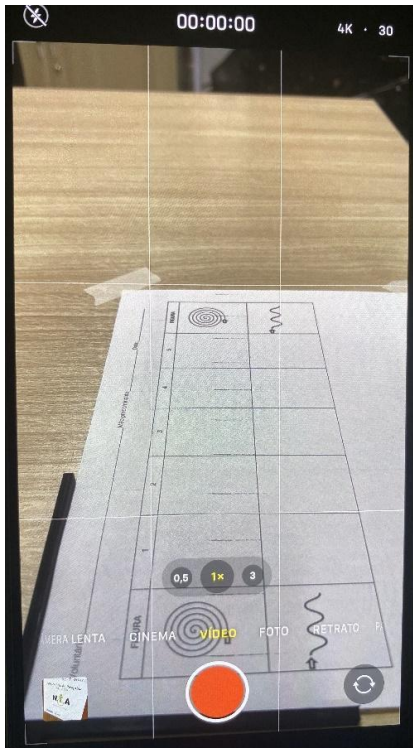


Figure 4. Camera field of view

C. *Data preprocessing and quality*

After collecting the two videos, it's time to certify the quality of the videos before collecting the next volunteer's data. It is necessary to certify if the videos are clear and have captured the pen tip during the drawing process. If the videos are not good enough, it is necessary to repeat the process until they are captured in good quality.

Considering the videos are good, it's necessary to process them in video editing software. It should be synced at the beginning of the video with the exact beginning of the drawing as well as the end. Then the audio must be excluded. The videos are stored in a folder with the date of the collecting process to keep them organized and accessible.

Once the video editing is finished and stored, the data is now ready to go to the DeepLabCut software and process the video there.

D. *DeepLabCut processing*

The videos must be inserted and processed one at a time in the DeepLabCut software. The first step is to load a pre-trained network by the authors and then go to the "Analyze Videos" tab and insert the configurations of the videos (e.g., mp4 or MOV format). After setting it all up, it's necessary to click the "RUN" button and wait until the software shows up the graphics. Now the graphics and the ".csv" file containing all the data about the video are saved in the same folder as the video.

The next step is to go to the "Create Video" tab, select the same video from the earlier step, set the right configurations for the videos, and click "Start.". Now the DLC software will create a new video, but this new video has a dot right at the pen tip. Which is our desired object to be tracked.

In order to maintain good quality and organization, it is not needed to move the videos because they are created and maintained in the same folder as the original video as well as the new data.

III. DISCUSSION

The protocol was based on studies that have worked with computer vision and produced good-quality video and results. This inspired the actual protocol decisions and facilitated their creation. The protocol is working well, considering a test made with the principal author as a volunteer to draw the test, but it should be tested with a pilot test, which will validate if the process is working, has good quality, and has reliable replication. Therefore, the protocol may have to undergo changes after the pilot test. The process nowadays is simple and fast to set up, and these qualities are very important to keep the collecting process flowing smoothly with no problems.

IV. CONCLUSION

The protocol has cameras, objects, field of view, and environmental issues under control. These are some of the concerns shown in some reference studies. Therefore, This is a simple project that can achieve excellent results, is fast to set up, and has good quality on a dimensional field. Now it is needed to do a pilot study to validate this protocol and search for points to be corrected and new improvements to be made.

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